Experiment No : 3

Date of Performance:
Date of Submission:

Aim : Demonstration of frequency reuse concept in cellular systems (simulation/visualization).

Required Equipment / Software :

- MATLAB (R2020a or later recommended).
- A computer with Windows/Linux/macOS.
- Basic knowledge of MATLAB scripting and plotting.

Theory :

A cellular communication system divides a geographical service area into a number of smaller regions called cells. Each cell is served by a base station and is allocated a set of radio frequency channels. Since the total available radio spectrum is limited, the same frequency channels can be reused in cells that are sufficiently separated. This technique is called frequency reuse.

The main purpose of frequency reuse is to increase the capacity of a cellular network without requiring additional spectrum. Cells using the same frequency set are called co-channel cells. They must be separated by an adequate distance so that co-channel interference remains within acceptable limits.

Frequency Reuse Ratio: $Q = D/R$

where D is the distance between the centers of the nearest co-channel cells and R is the cell radius.

For a hexagonal cellular system: $D/R = \sqrt{(3N)}$

where N is the cluster size (number of cells in one complete frequency-reuse cluster). The cluster size is commonly represented by $N = i^2 + ij + j^2$, where i and j are non-negative integers. Typical cluster sizes include 3, 4, 7, 9, 12, etc.

For example, for N = 7, the available frequency channels are divided into seven groups. Each group is assigned to one cell in a seven-cell cluster. The same group is then reused in other cells located at the appropriate reuse distance. A smaller N gives higher capacity but generally increases co-channel interference, whereas a larger N increases the reuse distance and reduces interference at the cost of lower frequency reuse efficiency.

Procedure :

1. Open MATLAB and create a new script file.

2. Define the cell radius R and select the cluster size N. For the main demonstration, use N = 7.

3. Generate the centers of hexagonal cells over a two-dimensional grid.

4. Plot each cell as a regular hexagon using MATLAB plotting functions.
5. Assign frequency groups F1 to F7 to the cells in the selected reuse pattern.

6. Repeat the frequency assignment over the complete network so that identical frequency groups occur in separated cells.

7. Display the frequency-group labels at the center of each hexagonal cell.

8. Use different line styles or markers for clarity and add a legend indicating the frequency groups.

9. Calculate the theoretical reuse ratio $Q = \sqrt{(3N)}$. For $N = 7$, $Q \approx 4.58$.

10. Run the MATLAB script and observe the repeated frequency pattern.

11. Change N, if desired, and compare how the reuse distance and frequency assignment

change. 12. Record the observations in the observation table.

MATLAB Program

```
clc;
clear;
close all;

% Frequency reuse demonstration
R = 1; % Normalized cell radius
N = 7; % Cluster size

% Frequency groups
freq = 1:N;

% Hexagonal cell geometry
theta = (0:6)*pi/3 + pi/6;
hx = R*cos(theta);
hy = R*sin(theta);

% Generate a cellular grid
xIndex = -3:3;
yIndex = -3:3;

figure('Color','w');
hold on;
axis equal;
grid on;
box on;

count = 0;
```

```
for m = xIndex
        for n = yIndex
% Center coordinates of pointy-top hexagons  xc =
1.5*R*m;
yc = sqrt(3)*R*(n + 0.5*mod(m,2));

% Skip cells outside a circular display area
if sqrt(xc^2 + yc^2) > 5.5*R
continue;
end

count = count + 1;

% Assign frequency group using a repeating pattern  group
= mod(m + 2*n, N) + 1;

% Draw hexagonal cell
patch(xc + hx, yc + hy, 'w', ...
         'EdgeColor', 'k', 'LineWidth', 1.2);

% Label the frequency group
text(xc, yc, sprintf('F%d', group), ...
'HorizontalAlignment', 'center', ...
'FontWeight', 'bold', 'FontSize', 10);
        end
end

% Display reuse information
Q = sqrt(3*N);
D = Q*R;

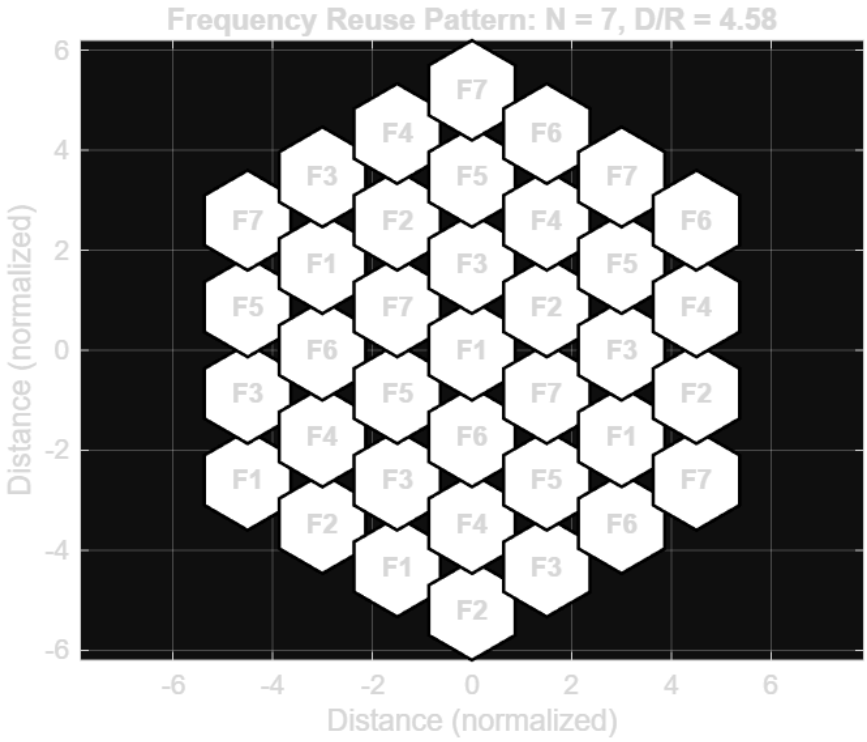
title(sprintf('Frequency Reuse Pattern: N = %d, D/R = %.2f', N, Q));
xlabel('Distance (normalized)');
ylabel('Distance (normalized)');

fprintf('Cluster size N = %d\n', N);
fprintf('Normalized cell radius R = %.2f\n', R);
fprintf('Reuse ratio D/R = %.4f\n', Q);
fprintf('Normalized reuse distance D = %.4f\n', D);
fprintf('Number of displayed cells = %d\n', count);

hold off;
```

Observation Table :

Sr. No.	Cluster Size (N)	Cell Radius R	Reuse Ratio Q = √(3N)	Frequency Groups	Observation
1	3	1.00	3.0000	3	A smaller cluster gives higher frequency reuse but greater co-channel interference.
2	4	1.00	3.4641	4	Reuse distance increases compared with N=3, reducing interference
3	7	1.00	4.5826	7	Good balance between network capacity and co-channel interference.
4	9	1.00	5.1962	9	Larger reuse distance reduces interference but decreases frequency reuse efficiency.



Result :

The frequency reuse concept in a cellular system was successfully demonstrated using MATLAB. A hexagonal cellular layout was generated and frequency groups were assigned according to a reuse pattern. For the selected cluster size N = 7, the theoretical reuse ratio was calculated as $Q = \sqrt{(21)} \approx 4.58$. The visualization clearly showed that the same frequency group can be reused in suitably separated cells.

Conclusion:

The experiment demonstrates how frequency reuse improves spectrum utilization in cellular communication systems. By dividing the service area into cells and reusing the same frequency groups at adequate separation distances, a cellular network can support more users with a limited amount of spectrum. MATLAB provides an effective platform for visualizing the hexagonal cellular structure, frequency allocation, cluster size, and reuse distance.

Viva Voce Questions :

1. What is frequency reuse?
2. Why is frequency reuse required?
3. What is a co-channel cell?
4. What is cluster size N?
5. What is the reuse distance?

MARKS & SIGNATURE :

R1 (3 Marks)	R2 (5 Marks)	R3 (4 Marks)	R4 (3 Marks)	Total (15 Marks)	Signature